

Total No. of Questions : 6]

SEAT No. :

P61

[Total No. of Pages : 2

TE/INSEM/APR-68

T.E. Information Technology (Semester - II)

314452 : MULTIMEDIA TECHNOLOGIES

(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicates full marks.
- 4) Assume Suitable data if necessary.

Q1) a) State and explain various components of multimedia. Also, explain the need of multimedia. **[5]**

b) Write and explain Huffman Coding algorithm for the input with frequencies of occurrences A(3) B(5) E(8) H(2) I(7). Construct Huffman Tree. **[5]**

OR

Q2) a) Define Multimedia. Why is multimedia important? Explain any two web applications. **[5]**

b) Explain Arithmetic coding Algorithm. Encode the message a1a2a3a4 with initial probabilities of $a_1 = 0.2$, $a_2 = 0.2$, $a_3 = 0.4$, $a_4 = 0.2$ **[5]**

Q3) a) Explain smoothing and sharpening filters used for image enhancement. **[6]**

b) Write a short note on color CMY and HSV color model. **[4]**

OR

Q4) a) Explain Vector Quantization lossy image compression. **[5]**

b) Explain JPEG image compression with neat sketch. **[5]**

P.T.O.

- Q5) a)** Write and explain MPEG Audio coding algorithm. **[5]**
b) What is MIDI? Explain MIDI with MIDI file structure and MIDI sound attribute. **[5]**

OR

- Q6) a)** Explain Predictive coding and delta modulation audio compression technique. **[6]**
b) Explain. WAV, WMA, .MP3 Audio file formats. **[4]**

